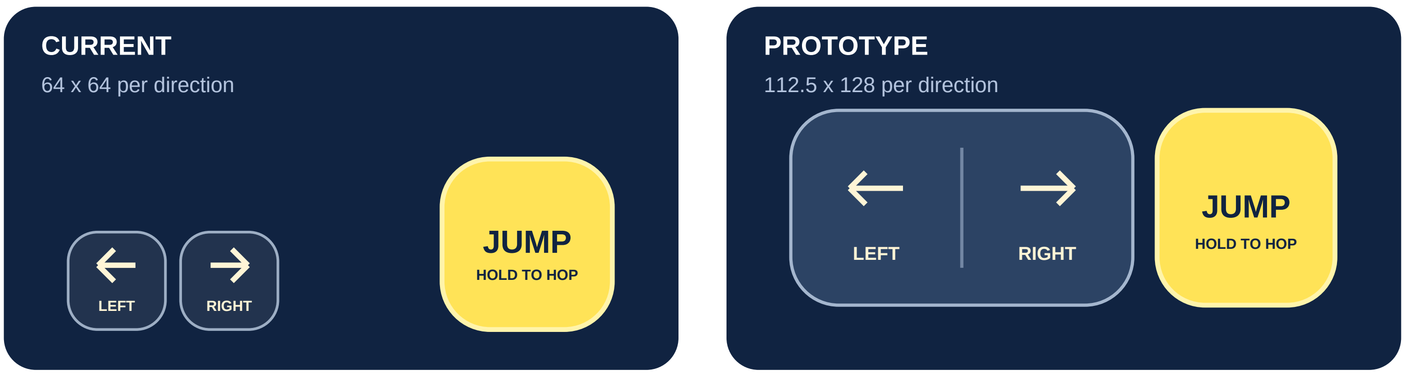
Make steering easy to find

Recommend a broad, two-part steering pad under the left thumb and a separate Jump pad under the right. A tap changes one lane immediately. Sliding across the divider changes direction without lifting. Holding a side does not repeatedly change lanes. Jump keeps its existing hold-to-hop and release/re-tap double-jump behavior.

The current controls measure 64 x 64 CSS pixels, with a 10-pixel gap and a minimum 16-pixel bottom inset. Those are code measurements, not measured physical dimensions on an iPhone. A near-miss outside either button is ignored. At a 390-pixel-wide viewport without side safe insets, the prototype gives each direction a 112.5 x 128 CSS-pixel target: approximately 3.5 times the area. Both controls sit higher, within the space already reserved below the car.



Control-area schematic at 390 CSS pixels wide. Not a browser screenshot or a physical-size measurement.

What the evidence supports

W3C says its 44 CSS-pixel enhanced target criterion is a minimum and recommends larger targets for frequent, sequential, hard-to-reach or edge-positioned controls. Passing a minimum does not establish gaming comfort or overall accessibility. [W3C: Target Size (Enhanced), updated May 11, 2026].

Apple's game guidance recommends comfortable thumb placement, safe-area clearance, large movement-input regions and visible press feedback. The prototype applies these principles with distinct direction halves, a clear divider and a highlight visible around the finger. Native UIKit points and browser CSS pixels are not interchangeable physical measurements. [Apple: Game controls, accessed September 5, 2026].

Design decision

Keep a visible, predictable direction pad as the first prototype. It directly addresses missed targets while retaining immediate taps for Autoroo's narrow timing windows. The target dimensions, higher placement and 12-pixel reversal guard are design choices to evaluate, not scientifically proven optimal values.

What comparable browser games do

These are verified mobile mappings from the platforms that host the games. They are documentation evidence, not hands-on playtests or evidence that a control scheme caused a game's success.

Game / source	Documented mobile controls	Relevance to Autoroo
Subway Surfers / Poki	Swipe sideways to move, up to jump, down to roll; double tap for hoverboard. Updated Aug 2026.	Closest lane-runner example for gestures. No hold-to-hop mapping documented.
Temple Run 2 / Poki	Swipe sideways to turn/change lanes, up to jump, down to slide. Updated Oct 2023.	Independent runner example. This is the browser mapping, not assumed native tilt behavior.
Level Devil / Poki	On-screen directional arrows plus a separate jump button. Updated Aug 2026.	Closest simple direction-plus-jump button precedent. Exact target sizes are undocumented.
Drive Mad / Poki	Tap and hold arrows for forward/backward motion. Updated Jun 2026.	Shows sustained touch controls in a browser driving game; no separate jump action.
Obby: Parkour with Ragdoll / CrazyGames	Left joystick; right buttons for jump, crouch and punch. Released Jul 2026.	Supports separate movement/action regions. Landscape and analog movement differ from Autoroo.
Big Catch / CrazyGames	Left joystick, right jump button; swipes control camera. Updated Feb 12, 2026.	Demonstrates why gesture regions need clear ownership. A weaker, landscape exploration analogy.

All six sources accessed September 5, 2026. Listed dates describe game updates/releases, not independently dated control documentation. No verified listing establishes a better miss rate, latency or target dimension for Autoroo.

Why not replace everything with swipes?

Swipes are a credible next alternative, particularly for one-handed play. However, interpreting a direction requires finger travel, and replacing Jump would also need a new mapping for its held and re-tapped states. Keep that as a separately evaluated option rather than combining competing tap and swipe meanings on the same surface.

A floating joystick accepts varied starting positions, but continuous deflection is a less direct fit for discrete lane steps. An original gamepad study found directional buttons precise for direction-restricted tasks but attention-demanding; floating controls reduced glances, with drift tradeoffs. It used a phone to control a separate display, limiting transfer to this game. Baldauf et al., ACM TOMM, 2015; TU Wien record.

The interaction contract

- Every new tap requests one lane immediately. A deliberate crossing requests the opposite lane. A small movement near the divider does not repeatedly steer the car.
- Jump has its own pointer. Steering remains possible while Jump is held, including during airborne lane changes. A steering finger moving near Jump never becomes a Jump press.
- Capture keeps tracking a contact after it leaves the original button. Cancel, lost capture, hiding the page, focus loss and resizing release held state. Controls clear on pause/unmount.
- The two direction targets meet at the divider without overlapping. The outer target geometry stays fixed while press feedback changes inside it. No extra gameplay banners are added.

Pointer capture and pointer identity support continued multi-touch tracking. Browsers may cancel a pointer when handling their own gesture; touch-action declares control-region intent. These are implementation safeguards, not a promise that a website can suppress iOS system-edge gestures. MDN: Pointer events and MDN: touch-action, accessed September 5, 2026.

Timing-critical gameplay retains activation on press-down. W3C's cancellation guidance allows essential down-event timing while recommending release/cancellation for ordinary controls. Menus keep normal button behavior. W3C: Pointer Cancellation, accessed September 5, 2026.

Evidence limits and review on your iPhone

An original 20-participant right-thumb study found larger targets faster and bottom-row taps biased above target centers. It used an older handheld and lift-off activation; it cannot prescribe the best location for modern two-thumb gaming. Parhi, Karlson and Bederson: Target Size Study, MobileHCI 2006.

The correct next step is actual-device comparison. On the same phone and grip, compare short runs using the current public version and the local prototype. Try rapid left/right changes, repeated same-direction taps, steering while holding Jump, re-tapping for double jump, pause/resume and rotation. Note missed presses, unintended reversals, occasions you look away from the road and thumb fatigue. Score alone is confounded by learning and random traffic.

What has been verified

42 focused automated tests passed across the new pad, existing mobile input, keyboard/persistence and camera suites. They cover divider/corner taps, centre jitter, no hold-repeat, cancellation, independent Jump and the production lane-roll/jump path. Type checking, lint and the static build passed. Static layout calculations fit 320-pixel portrait and 568 x 320 landscape; the mobile camera and desktop control paths were not changed.

No browser interaction test or physical iPhone playtest was performed. Research supports this prototype's direction; it does not prove a universal best layout or a measured improvement in miss rate. The game remains local for review. Nothing was pushed or deployed.